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# Two-step hybrid modeling for variable selection and estimation: An application to quantitative structure activity relationship study

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**Abstract**

In this study, we developed a simple technique for effective parameter estimation and prediction of the quantitative structure activity relationship studies using a two-step procedure. The first step is to choose the important molecular descriptors using the random forest regression, and the second step is to optimally predict the biological activity of the selected chemical compounds using the following estimators: ridge regression, jackknife ridge, Liu regression, jackknife Liu, Kibria–Lukman, and jackknife Kibria–Lukman. We conducted a simulation study and a real-life analysis with a quantitative structure–activity relationship (QSAR) data with 2540 descriptors after preprocessing. The optimal prediction is determined using the cross-validation error. The estimator with minimum cross-validation error is considered best. It is obvious that performing jackknife estimation after random forest selection is preferred.

**KEYWORDS**

cross-validation, jackknife, molecular descriptors, random forest, variable selection

## 1 | INTRODUCTION

Indexing of high-dimensional (HD) data is becoming increasingly important in many applications. HD data refers to situations when the number of features (predictors/explanatory variables) ($p$) exceeds the number of observation ($n$). HD data have become more frequent in different scientific fields such as genomics, medical sciences, biological sciences, computational chemistry, and others due to newly automated data collection techniques and technology advancement. For instance, in HD cancer classification, there is a need to select the important genes because the number of genes ($p$) exceeds the number of patients ($n$).[1,2] Also, high-dimensionality occurs in quantitative structure–activity relationship (QSAR) studies where the number of molecular descriptors is more than the number of compounds.[3,4] HD data pose threat to statistical analysis as most conventional methods of estimation in low-dimensional data/model are no longer available. For instance, the method of ordinary least squares (OLS) is the default method for fitting linear models, but is not appropriate for fitting HD regression model (applicable for problems with dimensionality larger than the sample size). Data visualization in HD datasets is extremely difficult. Associated problems in HD regression model include multicollinearity, redundancy, over-fitting, noisy features, and interpretation.[5,6]

There is poor prediction and interpretation difficulty due to the presence of high variance when linear models are fitted to HD data. Dimension reduction and variable selection methods have made immense progress in this regard. Examples of dimension reduction techniques include principal component analysis, factor analysis, and others. Variable selection methods enhance the interpretation of the estimated model by eliminating irrelevant features. Different

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variable selection methods have been developed over the years. These include the classical approach, the penalized estimators, genetic algorithms, particle swarm optimization, and others.[7–10] Most classical approaches such as forward selection, backward elimination, and stepwise regression techniques are based on the method of least squares. However, the method of least squares fails to give unique results when dealing with HD data.

Penalized estimators such as the least absolute shrinkage and selection operator (LASSO), the smoothly clipped absolute deviation (SCAD), or adaptive LASSO, elastic net (Enet), the Genc and Ozkale estimator (GO) estimator, the Liu-LASSO, Scaled-LASSO (SLASSO), and others are mostly employed for sparse and HD estimations.[11–17]

Recent studies have shown the advantage of combining the dimension reduction approach and the penalized estimators or combining two or more feature selection methods resulting to a form of hybrid/embedded estimators. They claimed that the hybrid estimators retain the advantages of the combined methods and dampen their drawbacks.[18–21] Recently, Arashi et al.[21] adopted the random forest regression for descriptor selection and employed the ridge or the Liu estimator for prediction. A recent study shows that the jackknife Kibria–Lukman (JKL) estimator outperforms the ridge or the Liu estimator when there is multicollinearity.[22]

Thus, in this paper, we propose a hybrid method by combining random forest regression and a few jackknifed estimators. At first, random forest regression is employed to select the important variables such that the dataset becomes low dimensional ($p < n$), and then prediction is conducted using the jackknife estimator. We conduct simulation and application studies to investigate the superiority of this hybrid method over some existing ones. The other part of this manuscript is structured as follows: Section 2 gives the theoretical background and estimation techniques for both low-dimensional and HD regression model when multicollinearity exists. In Section 3, we will outline the steps for the propose technique. Section 4 is devoted for the QSAR model prediction with 2540 descriptors and 134 observations while we conducted a simulation study in Section 5. Concluding remarks are provided in Section 6 with future recommendations.

## 2 | THEORETICAL BACKGROUND

Consider the following linear regression model:

$$\underline{\boldsymbol{y}}_i = \underline{\boldsymbol{x}}_i \underline{\boldsymbol{\beta}} + \underline{\epsilon}_i, \tag{1}$$

where the response variable $\underline{\boldsymbol{y}}_i$ is a $n \times 1$ vector, the predictors $\underline{\boldsymbol{x}}_i \in \mathbb{R}^{n\times p}$, $i = 1, ..., n$, is a known $n \times p$ matrix of predictors such that $p > n$, the regression coefficient $\underline{\boldsymbol{\beta}}$ is a $p \times 1$ vector, and $\underline{\epsilon}_i$, is the random error vector assumed to be normally distributed with mean 0 and variance $\boldsymbol{\sigma}^2 \in \mathbb{R}^+, i = 1, ..., n$. In low dimensional settings when $p < n$, $\boldsymbol{\beta}$ is popularly estimated using the least squares estimator (OLS). OLS minimizes $\|\boldsymbol{y} - \boldsymbol{X\beta}\|_2^2$ subject to an L2 norm with respect to $\boldsymbol{\beta}$ but fails to give a unique estimate in high dimensional settings when $p > n$.[21] Another threat to the performance of OLS is multicollinearity, which surfaces as a result of the correlation or linear dependency among the predictors.[22–27] Biased estimators such as the ridge regression estimator,[28] the Liu estimator,[29] modified ridge-type estimator,[30] the Kibria–Lukman (KL) estimator,[31] robust principle component (PC)-ridge estimator,[24] JKL estimator,[22] and others were developed to account for multicollinearity problem in linear regression models. The ridge estimator[28] minimizes $\|\boldsymbol{y} - \boldsymbol{X\beta}\|_2^2$ subject to an L2-norm of the coefficients and is defined as follows:

$$\widehat{\boldsymbol{\beta}}_n^k = \arg \min_{\beta \in \mathbb{R}^p} \left\{ \|\boldsymbol{y} - \boldsymbol{X\beta}\|_2^2 + k\|\boldsymbol{\beta}\|_2^2 \right\} = \left(\boldsymbol{X}^T\boldsymbol{X} + kI_p\right)^{-1}\boldsymbol{X}^T\boldsymbol{y}, \tag{2}$$

where $k > 0$ is the tuning parameter and $I_p$ is the identity matrix with ... and is better than the OLS estimator. The ridge estimator does not shrink any of its coefficient to zero. Thus, it cannot perform variable selection. However, the estimator gives more stable result and prediction than the OLS.[28] It is more complicated to interpret the result in HD settings. The jackknife ridge regression is defined by

$$\widehat{\boldsymbol{\beta}}^{\mathrm{JR}} = \left(\boldsymbol{X}^T\boldsymbol{X} + \mathrm{k}I_p\right)^{-1}\boldsymbol{X}^T\boldsymbol{X}\widehat{\boldsymbol{\beta}}^{\mathrm{LS}} \tag{3}$$

where $k > 0$ is the shrinkage parameter and $\widehat{\beta}^{\mathrm{LS}}$ is the OLS estimator. The jackknife ridge regression reduces the bias in the ridge regression. Liu[29] defined the Liu estimator to address multicollinearity:

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$$\widehat{\boldsymbol{\beta}}_n^{Liu} = \left(\boldsymbol{X}^T\boldsymbol{X}+I_p\right)^{-1}\left(\boldsymbol{X}^T\boldsymbol{X}+dI_p\right)\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y}, \tag{4}$$

where $d>0$ is the tuning parameter. The estimator is preferred to the ridge estimator in handling multicollinearity because it is linear in the tuning parameter. The jackknifed version of the Liu is

$$\widehat{\boldsymbol{\beta}}^{\mathrm{JL}} = \left(I_p-\left(\boldsymbol{X}^T\boldsymbol{X}+I_p\right)^{-2}\left(I_p-\mathrm{d}\right)^2\right)\widehat{\boldsymbol{\beta}}^{\mathrm{LS}} \tag{5}$$

where $\mathbf{d}$ is the shrinkage parameter. The jackknifed version is better than the Liu estimator in terms of reducing the bias in the Liu estimator. The KL estimator[30] dominates the ridge and the Liu estimators and is defined as follows:

$$\widehat{\boldsymbol{\beta}}_n^{KL} = \left(\boldsymbol{X}^T\boldsymbol{X}+kI_p\right)^{-1}\left(\boldsymbol{X}^T\boldsymbol{X}-kI_p\right)\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y}, \tag{6}$$

where $k>0$ is the tuning parameter. Ugwuowo et al.[22] found that the KL estimator performs well but with high bias. They jackknife the estimator to reduce its bias and called it the JKL estimator. JKL estimator is defined as follows:

$$\widehat{\boldsymbol{\beta}}_n^{JKL} = \left[I_p-2k\left(\boldsymbol{X}^T\boldsymbol{X}+kI_p\right)^{-1}\right]^2\left[I+2k\left(\boldsymbol{X}^T\boldsymbol{X}+kI_p\right)^{-1}\right]\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{y} \tag{7}$$

Most of these biased estimators except ridge and modified ridge-type will not give a unique estimate in HD settings. However, recently they have been adopted as post estimation techniques with good performance.[21]

The penalized estimation techniques are examples of variable selection methods that select important variables and shrink the irrelevant ones to zero. Much attention has been given to study different penalized estimators. For instance, LASSO[1] is defined as

$$\widehat{\boldsymbol{\beta}}_n^{Lasso} = \arg\min_{\beta\in\mathbb{R}^p}\left\{\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2+\lambda\|\boldsymbol{\beta}\|_1\right\}, \tag{8}$$

where $\lambda>0$ is the shrinkage parameter. LASSO often chooses only one variable from the group of pair-wise correlated features.[13,17] Zou and Hastie[13] developed the Enet approach by combining the L1 norm of LASSO with the L2 norms penalty of ridge regression. This is given by

$$\widehat{\boldsymbol{\beta}}_n^{Enet} = \arg\min_{\beta\in\mathbb{R}^p}\left\{\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2+\lambda\left[\frac{\alpha}{2}\|\boldsymbol{\beta}\|_2^2+\frac{1-\alpha}{2}\|\boldsymbol{\beta}\|_1\right]\right\}, \tag{9}$$

where $\|\cdot\|_2$ denotes the L2-penalty and $\|\cdot\|_1$ is the L1-norm.

## 3 | PROPOSED METHODOLOGY

Random forest regression (RFR) is a supervised learning algorithm that performs regression and classification using an ensemble learning method. See Speiser et al.[33] for a comprehensive review of the RFR for classification and variable selection. RFR works by constructing a large number of decision trees during training and return the average prediction for the individual trees.[32–35] Random forests can be used to naturally rank the importance of variables in a regression or classification problem, and it is implemented in the R package randomForest or python library Scikit-learn (Sklearn).[36,37] Recently, Arashi et al.[21] selected important variables with the use of random forests. The high-dimensional data were reduced to low-dimensional data, and finally, the low-dimensional data were estimated using the Liu estimator. In this study, we will only select important variables using random forest. The method was preferred to LASSO and Enet. Therefore, we will limit the scope of this study to random forest selection and post selection. We adopted the JKL estimator[22] as alternatives to the Liu estimator after RFR. The pseudo codes (steps) for the proposed methods are as follows:

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Step 1: Select important variables by executing the RFR on the training set.
Step 2: Apply jackknife estimators such as jackknife ridge, jackknife Liu, JKL, and others (ridge, jackknife ridge, Liu, and KL) after step 1 for prediction.
Step 3: Compare the performance of the following estimators: RFR-OLS, RFR–ridge (RFR-R), RFR–jackknife ridge (RFR-JR), RFR–Liu (RFR-L), RFR–jackknife Liu (RFR-JL), RFR-KL, and RFR-JKL.
Step 4: Predict the fitted values using the test data.
Step 5: The estimator's performance is compared using the test mean squared error.

## 4 | SYNTHETIC DATA ANALYSIS

In this section, we conduct simulation studies to assess the performance of the high-dimension strategies using four different scenarios with standard structures.

The simulation model is as follows:

$$y = X\beta + \sigma\epsilon \tag{10}$$

**TABLE 1** Simulation results of RMSEP for Case 1 when $n = 50$, $p = 100$, and $\sigma = 1, 3, 5$ and Case 2 when $n = 50$, $p = 150$, and $\sigma = 1, 3, 5$.

| | | Case 1 | | | Case 2 | | |
|---|---|---|---|---|---|---|---|
| $\rho$ | Criterion | $\sigma = 1$ | $\sigma = 3$ | $\sigma = 5$ | $\sigma = 1$ | $\sigma = 3$ | $\sigma = 5$ |
| 0.8 | Ridge | 13.4088 | 20.1463 | 33.0446 | 272.7597 | 276.1552 | 292.1594 |
| | RFR-R | 1.2295 | 9.0782 | 28.8896 | 6.5539 | 17.8076 | 43.9587 |
| | RFR-JR | 1.2252 | 9.0421 | 28.1643 | 6.5375 | 17.6724 | 43.6463 |
| | RFR-OLS | 1.2658 | 9.5331 | 29.5466 | 6.7121 | 18.6340 | 45.6778 |
| | RFR-L | 1.2436 | 9.3726 | 29.4172 | 6.5785 | 18.1004 | 44.2278 |
| | RFR-JL | 1.2428 | 9.3483 | 29.3396 | 6.5573 | 18.0530 | 44.0742 |
| | RFR-KL | 1.2123 | 9.0180 | 29.0412 | 6.4561 | 17.3429 | 43.3037 |
| | RFR-JKL | **1.2011** | **9.0038** | **28.9798** | **6.3042** | **17.1504** | **43.2649** |
| 0.9 | Ridge | 7.7419 | 15.4202 | 37.8193 | 505.4214 | 520.3106 | 521.8415 |
| | RFR-R | 1.4021 | 9.4772 | 29.0454 | 53.8635 | 61.4822 | 72.1769 |
| | RFR-JR | 1.3973 | 9.2287 | 27.8755 | 52.4809 | 59.6906 | 70.3316 |
| | RFR-OLS | 1.4528 | 10.2309 | 32.7912 | 58.6329 | 66.7701 | 77.8732 |
| | RFR-L | 1.3998 | 9.7727 | 31.0069 | 54.6153 | 62.5924 | 73.1610 |
| | RFR-JL | 1.3876 | 9.6757 | 30.3564 | 54.5670 | 62.1888 | 72.3597 |
| | RFR-KL | 1.3830 | 9.1084 | 26.9050 | 51.1432 | 59.0267 | 70.8007 |
| | RFR-JKL | **1.3805** | **9.1001** | **26.7713** | **51.0020** | **58.9458** | **70.1864** |
| 0.99 | Ridge | 9.2888 | 15.1857 | 41.1781 | 83.7128 | 91.5096 | 107.2902 |
| | RFR-R | 1.1444 | 9.7895 | 35.9918 | 1.5120 | 11.0372 | 31.2917 |
| | RFR-JR | 1.1001 | 9.5371 | 35.4286 | 1.4056 | 10.9180 | 29.8421 |
| | RFR-OLS | 1.4718 | 12.1178 | 39.7381 | 1.9380 | 14.7156 | 45.5633 |
| | RFR-L | 1.0924 | 9.5553 | 34.2105 | 1.2948 | 10.1848 | 27.5120 |
| | RFR-JL | 1.1790 | 9.5327 | 34.0200 | 1.2881 | 10.0039 | 27.3508 |
| | RFR-KL | 1.0832 | 9.6829 | 35.4407 | 1.3014 | 10.9848 | 29.6786 |
| | RFR-JKL | **1.0561** | **9.4218** | **35.3792** | **1.2906** | **10.8837** | **28.9470** |

*Note*: The best performing model in values is presented in bold.
Abbreviations: RFR-JKL, random forest regression–jackknife Kibria–Lukman; RFR-JL, random forest regression–jackknife Liu; RFR-JR, random forest regression–jackknife ridge; RFR-KL, random forest regression–Kibria–Lukman; RFR-L, random forest regression–Liu; RFR-OLS, random forest regression–ordinary least squares; RFR-R, random forest regression–ridge.

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where the random error $\epsilon \sim N(0, I)$. The training data were fitted using the glmnet, random forest, and other libraries in R (version 22—R Foundation for Statistical Computing).[38] The experiment was replicated 50 times with $p = 100$, 150, and 120; $\rho = 0.8$, 0.9, and 0.99. The cases are as follows:

Case 1: $n = 50$ (sample sizes), $p = 100$ (explanatory variables), $\sigma = 1, 3, 5$ (error standard deviation), $\sum_{kj} = \rho^{|k-j|}$, $j = 1, \ldots p$, such that $\rho = 0.8$, 0.9, and 0.99 (correlation), and $\boldsymbol{\beta} = \left(\mathbf{1}_5^T, \mathbf{0}_{p-5}^T\right)^T$ with 50 replications.[21,39]

Case 2: $n = 50$, $p = 150$, $\sigma = 1, 3, 5$, $\sum_{kj} = \rho^{|k-j|}$, such that $\rho = 0.8$, 0.9, and 0.99, and $\boldsymbol{\beta} = \left(\mathbf{3}_{10}^{\mathrm{T}}, \mathbf{0}_{\mathrm{p}-10}^{\mathrm{T}}\right)^{\mathrm{T}}$ replicated 50 times.

Case 3: $n = 50$, $p = 120$, and $\sigma = 3$. The explanatory variables are divided into m groups such that the design matrix is $\mathbf{X} = \{X_1, \ldots, X_{10}, \ldots, X_{31}, \ldots, X_{40}, X_{41}, \ldots, X_{120}\}$ where $\mathbf{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$. The diagonal elements of $\boldsymbol{\Sigma}$ are 1, and the non-diagonal elements within the group $(m = 1, 2, 3, 4)$ are $\rho$. The off-diagonal elements between any two groups are taken to be 0.2. The true regression coefficients are defined as follows: $\beta^{\mathrm{T}} = \left(\mathbf{5}_{10}^{\mathrm{T}}, \mathbf{5}_{-10,}^{\mathrm{T}} \mathbf{3}_{10}^{\mathrm{T}}, \mathbf{3}_{-10}^{\mathrm{T}}, \mathbf{0}_{\mathrm{p}-40}^{\mathrm{T}}\right)^{\mathrm{T}}$.

The model's performance is measured by prediction accuracy, so we calculate the root mean squared error of prediction using Equation (11).

**TABLE 2** Simulation results of RMSEP for Case 3 when $n = 50$, $p = 120$, $\sigma = 3$.

| | | Case 3 | | |
|---|---|---|---|---|
| $\rho$ | **Criterion** | $\sigma = 1$ | $\sigma = 3$ | $\sigma = 5$ |
| 0.9 | Ridge | 188.3236 | 200.9073 | 214.1948 |
| | RFR-R | 61.1980 | 79.3696 | 109.1191 |
| | RFR-JR | 71.4826 | 88.9688 | 117.1106 |
| | RFR-OLS | 62.5613 | 87.0629 | 124.7877 |
| | RFR-L | 56.9484 | 75.5545 | 107.7709 |
| | RFR-JL | 54.7108 | 75.4598 | 106.9174 |
| | RFR-KL | 88.5956 | 104.5719 | 130.9282 |
| | RFR-JKL | **87.3611** | **103.3775** | **128.9153** |
| 0.95 | Ridge | 123.8456 | 130.0656 | 147.1946 |
| | RFR-R | 44.4480 | 56.0326 | 74.0812 |
| | RFR-JR | 45.8805 | 57.2239 | 76.9050 |
| | RFR-OLS | 48.5176 | 63.8341 | 87.4946 |
| | RFR-L | 44.2531 | 55.5213 | 73.5930 |
| | RFR-JL | 43.8647 | 55.5069 | 72.1461 |
| | RFR-KL | 47.3832 | 58.4487 | 78.6944 |
| | RFR-JKL | **46.2841** | **56.1074** | **75.0806** |
| 0.99 | Ridge | 141.0938 | 150.7323 | 163.4460 |
| | RFR-R | 14.1990 | 23.6935 | 47.5874 |
| | RFR-JR | 13.6010 | 23.5011 | 47.3777 |
| | RFR-OLS | 20.5506 | 40.5811 | 81.7462 |
| | RFR-L | 12.9493 | 20.8217 | 39.5406 |
| | RFR-JL | 12.9272 | 20.5422 | 38.1020 |
| | RFR-KL | 16.2373 | 24.5286 | 49.9424 |
| | RFR-JKL | **15.2149** | **23.8174** | **44.1910** |

*Note*: The best performing model in values is presented in bold.
Abbreviations: RFR-JKL, random forest regression–jackknife Kibria–Lukman; RFR-JL, random forest regression–jackknife Liu; RFR-JR, random forest regression–jackknife ridge; RFR-KL, random forest regression–Kibria–Lukman; RFR-L, random forest regression–Liu; RFR-OLS, random forest regression–ordinary least squares; RFR-R, random forest regression–ridge.

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$$\mathrm{RMSEP} = \frac{1}{n_{\mathrm{TEST}}} \sum_{i=1}^{n_{\mathrm{TEST}}} \left( x_i^T \beta - x_i^T \widehat{\beta} \right)^2 \tag{11}$$

An estimator is preferred when it has the lowest test mean squared error defined in Equation (11).

**TABLE 3** Parameter estimate after random forest selection.

| | Coef | RFR-OLS | RFR-R | RFR-JR | RFR-L | RFR-JL | RFR-KL | RFR-JKL |
|---|---|---|---|---|---|---|---|---|
| Selected descriptor | RMSEP | 0.5057 | 0.4094 | 0.3801 | 0.7272 | 0.5894 | 0.4798 | 0.3413 |
| AMW | $\widehat{\beta}_1$ | −2.9675 | 4.3152 | 0.4317 | 0.1436 | 0.1573 | −1.8072 | 6.5680 |
| nBT | $\widehat{\beta}_2$ | 2.5742 | 2.5331 | 2.6443 | 1.7933 | 2.3071 | 2.5956 | 2.7685 |
| nBO | $\widehat{\beta}_3$ | 0.0513 | 0.0449 | 0.0214 | 0.0413 | 0.0180 | 0.0441 | −0.0303 |
| RBF | $\widehat{\beta}_4$ | −0.0555 | −0.0475 | −0.0342 | 0.0227 | −0.0257 | −0.0475 | 0.0066 |
| SCBO | $\widehat{\beta}_5$ | 1.9520 | 2.5090 | 0.5004 | 0.3297 | 0.4314 | 1.5507 | −1.9837 |
| RBN | $\widehat{\beta}_6$ | 0.4387 | 0.1303 | 0.8641 | 0.3653 | 0.4430 | 0.6210 | 1.7222 |
| RFD | $\widehat{\beta}_7$ | 0.0258 | 0.0477 | −0.0114 | −0.0879 | −0.0442 | −0.0057 | −0.0879 |
| NNRS | $\widehat{\beta}_8$ | 5.7754 | 6.2321 | 0.1158 | 0.1818 | 0.2441 | 3.9123 | −10.0341 |
| HNar | $\widehat{\beta}_9$ | 2.0803 | 1.8606 | 1.7381 | 0.1741 | 0.2892 | 2.1320 | 1.8540 |
| Ram | $\widehat{\beta}_{10}$ | −0.2310 | −10.3563 | −0.2593 | −0.0070 | −0.0143 | −0.2999 | −0.4409 |
| LPRS | $\widehat{\beta}_{11}$ | 0.0210 | 0.0184 | −0.0231 | 0.0398 | −0.0045 | 0.0049 | −0.1010 |
| MDDD | $\widehat{\beta}_{12}$ | 2.1130 | 0.6688 | 2.1906 | 0.5176 | 0.8122 | 2.3508 | 2.8456 |
| UNIP | $\widehat{\beta}_{13}$ | −2.7563 | −3.9503 | 3.0737 | 0.2899 | 0.4626 | −0.2557 | 14.4404 |
| VAR | $\widehat{\beta}_{14}$ | 1.0956 | 1.9058 | −0.1587 | 0.4329 | 0.6479 | 0.5167 | −2.8520 |
| ICR | $\widehat{\beta}_{15}$ | 0.0834 | 0.0714 | 0.0779 | 0.1533 | 0.1326 | 0.0783 | 0.0546 |
| PW2 | $\widehat{\beta}_{16}$ | 3.7286 | 2.5947 | 1.0573 | 0.3306 | 0.4322 | 3.0206 | −3.1333 |
| PW5 | $\widehat{\beta}_{17}$ | 3.4331 | 3.3985 | 0.4443 | −0.1272 | −0.0585 | 2.4977 | −4.8159 |
| Psi_i_s | $\widehat{\beta}_{18}$ | 5.5721 | 6.5225 | −0.8943 | 0.2923 | 0.4878 | 2.9276 | −13.2387 |
| Psi_e_A | $\widehat{\beta}_{19}$ | −2.7623 | −11.1528 | −0.5464 | −0.0375 | −0.0881 | −2.1237 | 3.1647 |
| MWC01 | $\widehat{\beta}_{20}$ | 0.0253 | 0.0231 | 0.0246 | 0.0252 | 0.0291 | 0.0257 | 0.0228 |
| MWC05 | $\widehat{\beta}_{21}$ | −0.0033 | −0.0143 | −0.0063 | 0.0103 | −0.0026 | −0.0041 | −0.0124 |
| MWC10 | $\widehat{\beta}_{22}$ | −0.5714 | −0.4747 | −0.3933 | −0.1454 | −0.2779 | −0.5002 | −0.0635 |
| SRW04 | $\widehat{\beta}_{23}$ | −0.0499 | −0.1534 | 0.0036 | 0.0572 | 0.0468 | −0.0326 | 0.0827 |
| SRW07 | $\widehat{\beta}_{24}$ | 1.4784 | 1.3785 | 0.5471 | 0.2978 | 0.3353 | 1.2176 | −0.9569 |
| MPC01 | $\widehat{\beta}_{25}$ | −0.3809 | −0.2824 | 0.3976 | 0.0794 | 0.1405 | −0.1393 | 1.8289 |
| piPC03 | $\widehat{\beta}_{26}$ | 0.0538 | 0.0449 | 0.0769 | 0.0272 | 0.0446 | 0.0618 | 0.1185 |
| piPC10 | $\widehat{\beta}_{27}$ | 0.6361 | −0.5066 | 0.4344 | 0.4213 | 0.4269 | 0.6758 | 0.2268 |
| TWC | $\widehat{\beta}_{28}$ | −2.4701 | −1.7669 | −2.0380 | −0.0833 | −0.2843 | −2.4685 | −1.6899 |
| BID | $\widehat{\beta}_{29}$ | −1.9912 | −2.0878 | −0.5280 | 0.2533 | 0.2717 | −1.8783 | 1.4928 |
| SIC0 | $\widehat{\beta}_{30}$ | −2.0637 | −1.7988 | −0.1420 | 0.3132 | 0.4295 | −1.6682 | 2.5938 |
| VE1_A | $\widehat{\beta}_{31}$ | 0.2214 | 0.2648 | 0.0859 | −0.0351 | −0.0096 | 0.1760 | −0.1468 |
| SpPosA_A | $\widehat{\beta}_{32}$ | −2.9230 | −2.5085 | −1.1200 | −0.1185 | −0.2485 | −2.3452 | 1.7903 |

Abbreviations: RFR-JKL, random forest regression–jackknife Kibria–Lukman; RFR-JL, random forest regression–jackknife Liu; RFR-JR, random forest regression–jackknife ridge; RFR-KL, random forest regression–Kibria–Lukman; RFR-L, random forest regression–Liu; RFR-OLS, random forest regression–ordinary least squares; RFR-R, random forest regression–ridge.

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The result of Case 1 and Case 2 is presented in Table 1 while that of Case 3 is in Table 2. It is obvious that performing ridge estimation only does not produce stable results as compared to the result obtained after selecting important variables with random forest. For instance, under Case 1, when $\sigma = 3$ and $\rho = 0.8$, the root mean squared error of prediction of ridge is 20.1463, which is higher than RFR-OLS. RFR-OLS is next in terms of low performance because there is tendency of multicollinearity in the model. The jackknife version of the RFR-R, RFR-KL, and RFR-L produced lower root mean squared error of prediction (RMSEP) as opposed to their counterpart. The results in Case 1 to Case 3 show that RFR-JKL dominates more while RFR-JR and RFR-JL compete favorably. It is obvious that the RMSEP increases as sigma increases from Case 1 to Case 3. However, in Case 3, RMSEP declines as the level of correlation increases. RFR-L performs better than RFR-R especially in Case 3 with few exceptions in Cases 1 and 2. This agrees with the study by Arashi et al.[21]

## 5 | QSAR DATA

QSAR is a chemistry data. Algamal and Asar[40] stated that the response variable is the biological activities of imidazo [4,5-b]pyridine derivatives as an anticancer compounds, and the response variables in QSAR modeling are positively skewed. In chemometrics, QSAR is widely used data for modeling, and it has a great deal of importance. The linear regression model is one of the most important models for constructing the QSAR modeling by relating several biological activities over a collection of chemical compounds in terms of their structural properties with several molecular descriptors.[1]

The dataset was adopted to study the HD QSAR prediction of the anticancer potency of a series of imidazo [4,5-b] pyridine derivatives. There were 4885 molecular descriptors, including all 29 blocks based on the optimized molecular structures.[17–19] The number of molecular descriptors for the QSAR modeling reduces to 2540 descriptors after removing the zeros. We applied the existing estimators and the proposed to the QSAR prediction of the anticancer potency of imidazo [4,5-b] pyridine derivatives by adopting the validation approach discussed in Section 3.

Table 3 shows that 32 important variables were extracted from the list of available predictors. We estimate the selected variables using OLS, ridge, jackknife ridge, Liu, jackknife Liu, KL, and JKL. The results show that performing jackknife estimation after RFR variable selection produced better result in terms of the RMSEP. The result of the real-life analysis agrees with the simulation study.

## 6 | CONCLUSION

In this study, we developed a simple technique for effective parameter estimation and prediction of the quantitative structure activity relationship studies using a two-step procedure. The first step is to choose the important molecular descriptors using the RFR, and the second step is to optimally predict the biological activity of the selected chemical compounds using the following estimators: ridge regression, jackknife ridge, Liu regression, jackknife Liu estimator, KL, and JKL. We conducted a simulation study exploring three different scenarios. The performance of the methods is judged using the root mean squared error of prediction. The results in Case 1 to Case 3 show that combining random forest with the jackknife estimators gives better prediction in terms of minimum root mean squared error of prediction. We adopted a QSAR data with initial descriptors of 4885 which was reduced to 2540 descriptors after preprocessing. The random forest regression was able to extract 32 important descriptors, and a second step estimation was performed for predictions. The result of the simulation study agrees with the application example.

### PEER REVIEW

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### DATA AVAILABILITY STATEMENT

Data will be available at the request by the reviewer by the corresponding author—Henrietta Ebele Oranye.

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